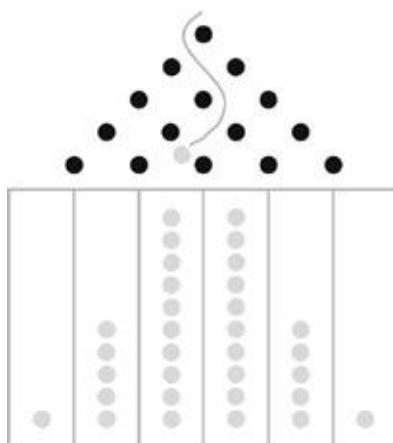


Assume that I set you up in a (legal) wager where the odds are neither in your favor nor against you. Flip a coin. Heads, you make \$1, tails, you lose \$1.

At the first flip, you will either win or lose.

At the second flip, the number of possible outcomes doubles. Case one: win, win. Case two: win, lose. Case three: lose, win. Case four: lose, lose. Each of these cases has equivalent odds, the combination of a single win and a single loss has an incidence twice as high because cases two and three, win-lose and lose-win, amount to the same outcome. And that is the key for the Gaussian. So much in the middle washes out—and we will see that there is a lot in the middle. So, if you are playing for \$1 a round, after two rounds you have a 25 percent chance of making or losing \$2, but a 50 percent chance of breaking even.

FIGURE 8: THE QUINCUNX (SIMPLIFIED)—A PINBALL MACHINE



Drop balls that, at every pin, randomly fall right or left. Above is the most probable scenario, which greatly resembles the bell curve (a.k.a. Gaussian distribution). *Courtesy of Alexander Taleb.*

Let us do another round. The third flip again doubles the number of cases, so we face eight possible outcomes. Case 1 (it was win, win in the second flip) branches out into win, win, win and win, win, lose. We add a win or lose to the end of each of the previous results. Case 2 branches out into win, lose, win and win, lose, lose. Case 3 branches out into lose, win, win and lose, win, lose. Case 4 branches out into lose, lose, win and lose, lose, lose.

We now have eight cases, all equally likely. Note that again you can group the middling outcomes where a win cancels out a loss. (In Galton's quincunx, situations where the ball falls left and then falls right, or vice versa, dominate so you end up with plenty in the middle.) The net, or cumulative, is the